BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 7 . MODULE 12 OF 16

Module 12: Lymphatic and Immune System

Topics: Lymphatic System and Innate Immunity, Adaptive Immunity

TOPIC 1 OF 2 . WEEK 7 . 5 CARDS (3 DOK 1 / 1 DOK 2 / 1 DOK 3)

Lymphatic System and Innate Immunity

Lymph vessels, lymphoid organs, and the body's general (non-specific) defenses.

The Science

The lymphatic system returns interstitial fluid to the blood (about 3 L per day) and filters that fluid through lymph nodes for immune surveillance. Lymph vessels parallel the venous system. Major lymphoid organs: lymph nodes (filter lymph), spleen (filters blood), thymus (T cell maturation), tonsils, and MALT (mucosa-associated lymphoid tissue scattered along the gut, airway, etc.).

Innate immunity is the body's general (non-specific) defense and is the first line. Surface barriers: skin (stratified squamous keratinized), mucous membranes (with antimicrobial peptides and IgA), low stomach pH, ciliated airway. If something gets past, innate cells respond: neutrophils (first responders, eat bacteria), macrophages (eat and recruit), natural killer cells (kill virus-infected and tumor cells without prior exposure). The complement system is a cascade of plasma proteins that lyse pathogens and mark them for phagocytosis. Interferons are antiviral cytokines.

Inflammation: the response to tissue damage or infection. Vasodilation and increased vascular permeability deliver immune cells and plasma proteins to the site. Cardinal signs: redness (rubor), heat (calor), swelling (tumor), pain (dolor), loss of function. Chemotaxis recruits more cells. Inflammation is necessary but can cause tissue damage if uncontrolled.

Teaching This Topic

Before the video (drawing prompt)

Have students sketch the path of interstitial fluid into a lymphatic capillary, through a lymph node, and back to the blood. Then list the five cardinal signs of inflammation.

Common misconceptions to address

- Students confuse the lymphatic and the cardiovascular systems. They are parallel but separate; lymph empties into the venous system at the subclavian veins.
- Innate immunity is sometimes dismissed as primitive. It handles 99% of insults without you noticing.
- Inflammation is seen as pathological. It is the body's defense; chronic or uncontrolled inflammation is the pathology.

Order of operations (what to teach first)

Lymphatic anatomy and function, then innate immunity layers (barriers, cells, complement), then inflammation as the visible result.

Self-test prompts

- Name the four cardinal signs of inflammation.
- Name two phagocytic cells.
- Why do asplenic patients need extra vaccines and antibiotic precautions?

Why This Matters to Your Patients

Lymphedema (often after lymph node dissection for breast cancer surgery, or from filariasis) is the failure of lymph drainage; the affected limb swells permanently. Splenectomy patients are at high risk of overwhelming infection from encapsulated organisms (Strep pneumoniae, H. influenzae, N. meningitidis); they receive specific vaccines. Sepsis is uncontrolled systemic inflammation in response to infection; early recognition (SIRS criteria, qSOFA, lactate) is critical for survival. Allergies and anaphylaxis involve inappropriate or excessive innate (and adaptive) responses. Nurses caring for any patient with infection are watching for the transition from local to systemic inflammatory response.

TOPIC 2 OF 2 . WEEK 7 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Adaptive Immunity

Specificity and memory: how B and T cells learn what to attack and remember it.

The Science

Adaptive immunity has two defining features: specificity (each lymphocyte recognizes one antigen) and memory (re-exposure produces a faster, larger response). It comes in two arms.

Humoral immunity: B cells, when activated, become plasma cells that produce antibodies. Antibodies (IgG, IgM, IgA, IgE, IgD) circulate and tag extracellular pathogens for destruction by neutralization, opsonization, complement activation, or antibody-dependent cell killing (ADCC). This is your primary defense against circulating bacteria, toxins, and viruses outside cells.

Cell-mediated immunity: T cells, in two main flavors. Helper T cells (CD4) coordinate the response by activating B cells, cytotoxic T cells, and macrophages. Cytotoxic T cells (CD8) directly kill infected or abnormal cells. T cells need antigen presented on MHC molecules: MHC class I (on all nucleated cells, presents intracellular peptides to CD8) or MHC class II (on antigen-presenting cells, presents extracellular peptides to CD4).

Activation requires two signals: antigen plus co-stimulation. This two-key rule prevents autoimmunity (a lymphocyte that binds self-antigen without co-stimulation is suppressed or killed). Clonal expansion produces effector cells (do the job) and memory cells (live for years, respond fast on re-exposure). Vaccination relies entirely on memory.

Teaching This Topic

Before the video (drawing prompt)

Have students predict: which arm of adaptive immunity is best at killing virus-infected cells versus extracellular bacteria? Then they have to commit.

Common misconceptions to address

- Students think innate and adaptive immunity work in series. They cooperate constantly; innate cells present antigen and drive adaptive responses.
- Antibodies are sometimes thought to kill pathogens directly. They tag, neutralize, or recruit; killing is usually by complement or phagocytes.
- Vaccines are sometimes thought to give you the disease. They give you the antigen without the pathogen; your memory cells will respond if you ever encounter the real one.

Order of operations (what to teach first)

Definition and features (specificity, memory), then humoral arm with antibody functions, then cell-mediated arm with MHC presentation, then activation rules and memory.

Self-test prompts

- Which T cell subset directly kills infected cells?
- Which MHC class presents to CD4 T cells?
- Why are vaccinated people protected even though they have never had the disease?

Why This Matters to Your Patients

HIV destroys CD4 T cells; without coordination, both humoral and cell-mediated immunity collapse, and patients develop opportunistic infections. Transplant patients receive immunosuppression to prevent T cell rejection of the graft; the trade-off is infection and cancer risk. Autoimmune diseases (lupus, rheumatoid arthritis, type 1 diabetes, MS, Hashimoto, Graves) reflect loss of self-tolerance. Vaccination has transformed pediatric mortality. Monoclonal antibody therapies (mAbs ending in -mab) target specific receptors or pathogens. Every patient with infection, autoimmunity, or immunodeficiency is a story about the adaptive immune system functioning, failing, or overshooting.